Lab Work 3: K-Means Clustering on Customer Sales Data

Objective: The purpose of this lab is to apply the K-Means clustering algorithm to categorize customers based on their sales and profit data. This will help the business understand customer behavior and segment them into meaningful groups for better decision-making.

Problem Statement: A retail company wants to divide its customer base into different segments based on total sales and profit. Using clustering, the company aims to identify unique customer groups, such as high spenders, low-profit customers, or potential loyal buyers.

Steps:

1. **Import necessary libraries.**

   Begin by importing necessary Python libraries such as:

```
# Step 1: Import libraries
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

2. **Load the dataset into a pandas Data Frame and perform data pre processing (if necessary).**
   - Load the dataset into a DataFrame using pandas.
   - Check for any missing values or inconsistencies.
   - Normalize or scale the data if required to ensure fair clustering.

```
# Step 2: Load data
df = pd.read_csv('Lab_3_Data.csv')

# Step 3: Select relevant features
data = df[['Sales', 'Profit']].dropna()

# Step 4: Scale data
scaler = StandardScaler()
data_scaled = scaler.fit_transform(data)
```

3. **Apply K-Means clustering to group the customers.**

- Select the relevant features: Sales and Profit.
- Choose a suitable number of clusters (e.g., k=3).
- Apply the K-Means algorithm to classify customers into clusters.

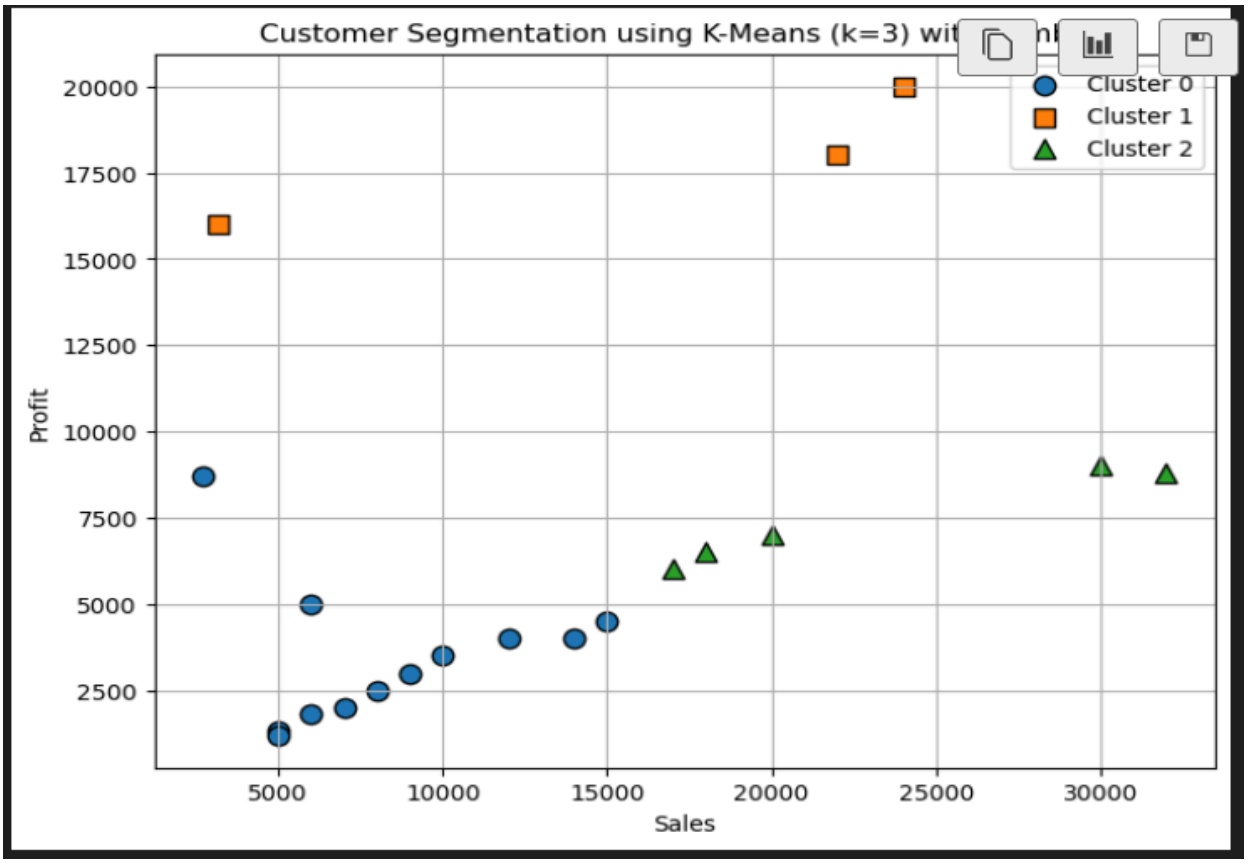
```
# Step 5: Apply KMeans
k = 3
kmeans = KMeans(n_clusters=k, random_state=42)
data['Cluster'] = kmeans.fit_predict(data_scaled)

# Step 6: Plot using different markers
markers = ['o', 's', '^', 'P', '*', 'X', 'D']  # Add more if k > 7
plt.figure(figsize=(8, 6))

for cluster in range(k):
    cluster_data = data[data['Cluster'] == cluster]
    plt.scatter(
        cluster_data['Sales'],
        cluster_data['Profit'],
        marker=markers[cluster % len(markers)],  # Use different symbols
        label=f'Cluster {cluster}',
        edgecolor='black',
        s=80  # point size
    )
```

4. **Visualize the clusters and interpret the output.**

- Use a scatter plot to visualize how customers are grouped.
- Assign different colors to each cluster for better clarity.



Expected Output:

1. **Scatter Plot Visualization:**

- Each customer is represented by a point.
- Different colours represent different clusters.
- The plot shows how customers are grouped based on sales and profit.

This visual helps in identifying groups like high sales-high profit, low sales-low profit, etc.

Discussion Questions:

How can the company use these clusters to improve customer engagement?

By targeting each cluster differently. For example:

- High profit cluster: Offer loyalty programs.

- Low profit cluster: Provide personalized promotions to boost sales.

What additional features could improve clustering accuracy?

Including more variables such as:

- Number of purchases.
- Time since last purchase.
- Customer demographics.
- Product categories bought.

What would happen if the number of clusters (k) is increased or decreased?

- Increasing k may over-segment the data, causing fragmentation.
- Decreasing k might oversimplify and combine dissimilar groups.
- The right value of k can be chosen using the Elbow Method.

Conclusion

This lab gives practical exposure to K-Means clustering, a key unsupervised learning technique. It helps in customer segmentation, which is essential for data-driven marketing and strategic planning.